

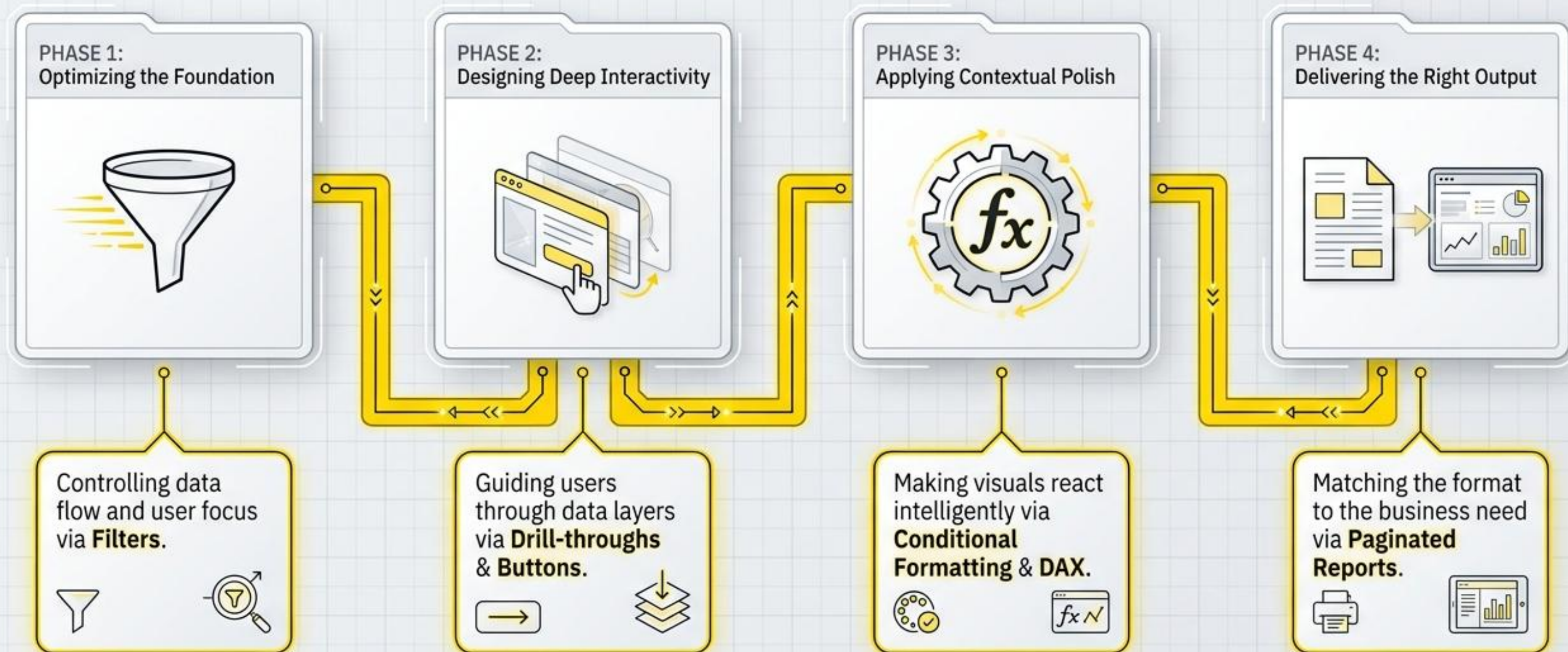
UI Card

POWER BI: ADVANCED DATA ANALYSIS AND VISUALIZATION

Architectural strategies for dynamic applications.
By dr Zbigniew Galar.

The Developer's Toolkit

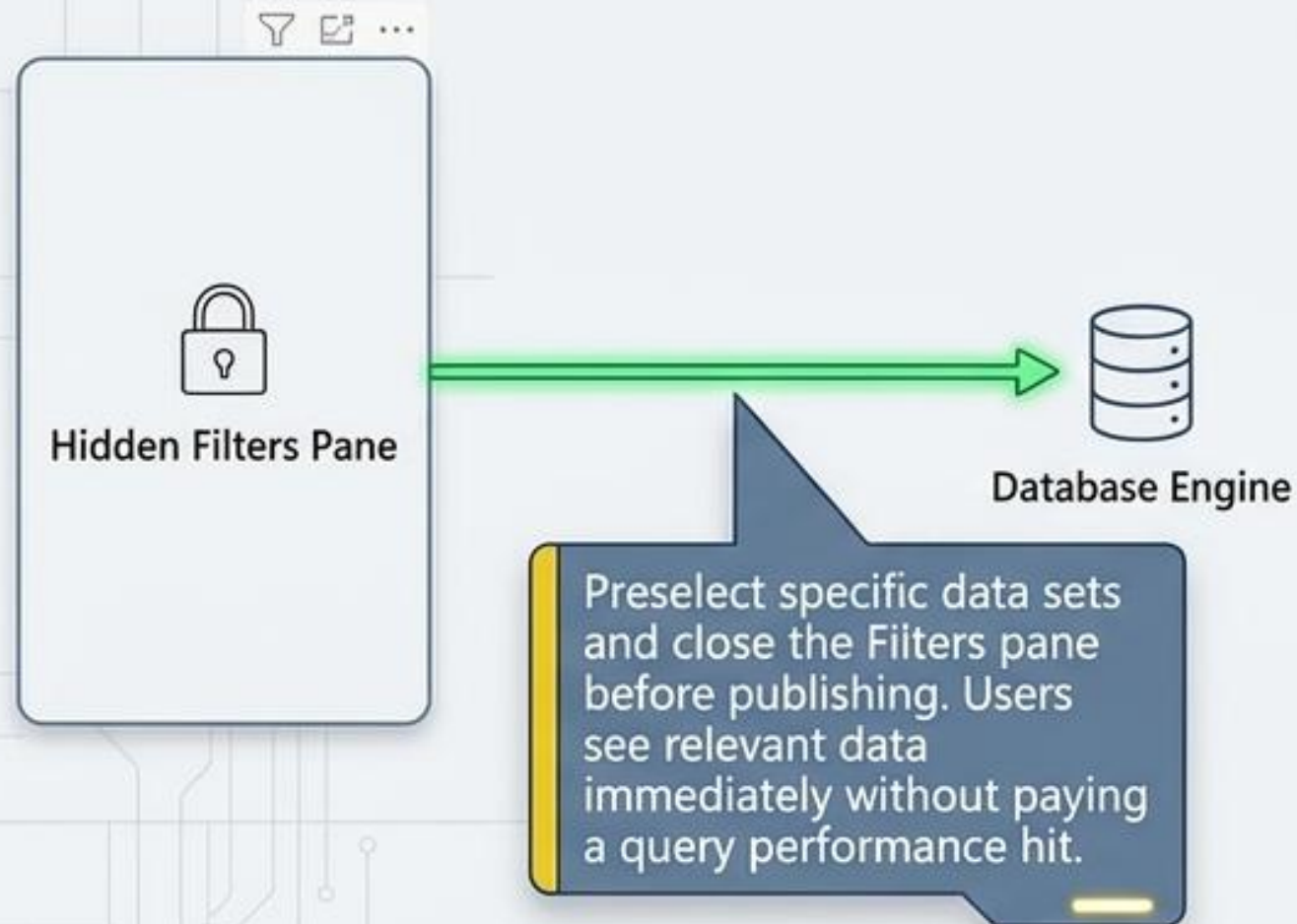
A Structural Blueprint for Interactive Design



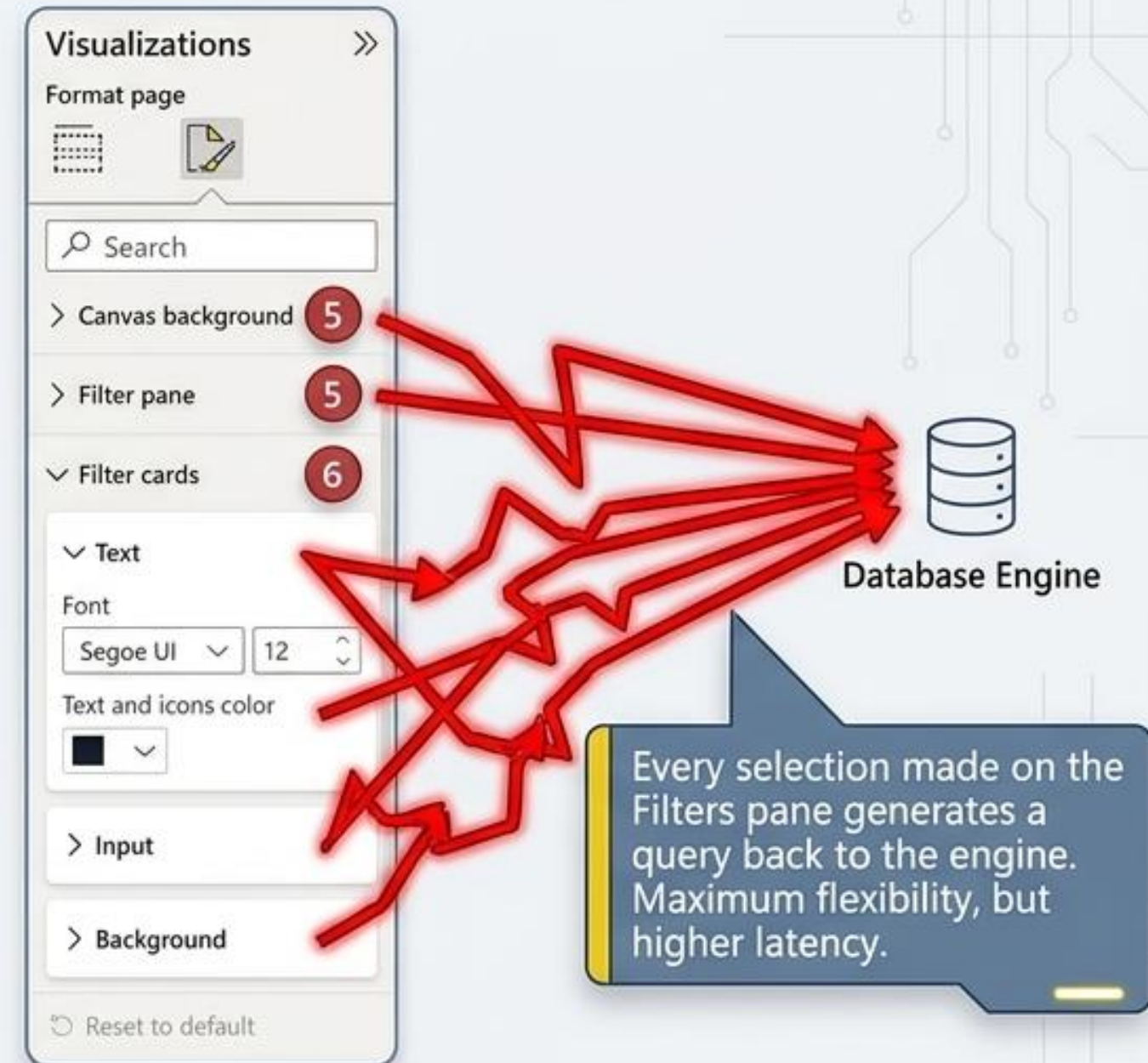
The Developer's Toolkit

Query Impact Flowchart: Comparing Developer Approaches

Optimized Approach: Pre-filtered & Closed Pane



User-Driven Approach: Open Pane / Dynamic



Design Rule: Match filter pane color schemes to the canvas background to create a continuous, app-like interface.

The Drill-Through Funnel



Origin: User right-clicks a specific data point (e.g., 'Meat/Poultry' category).

Filter Context Package

The selection 'carries through' as a filter context package to the destination page.

Destination: The page renders entirely filtered by the Origin selection. The Back Button appears, indicating successful setup.

Page B (Details)



Controlling the Drill-Through Environment

UI CONSOLE POWER

Control Panel 1: The Origin Slicer

Select all

(Blank)

1996

1997

1998

In addition to selecting a Category, this page also has a slicer with the year 1996 selected. This filter will also be transmitted.

ON

Keep All Filters

Toggle **OFF** to isolate the destination page and prevent origin filters from carrying over. Side benefit: Active slicer selections on the originating page transmit alongside the primary drill-through.

UI CONSOLE POWER

Control Panel 2: The Cross-Report Engine

Visualizations >>

Build visual

CategoryName

is (All)

Allow drill through when:

Used as category

Search

☒ Beverages

1

☒ Condiments

1

☐ Confections

1

☒ Dairy Products

1

☒ Grains/Cereals

1

☐ Meat/Poultry

1

☒ Produce

1

☒ Seafood

1

Drill through

Cross-report

Keep all filters

Off

On

The same drill-through architecture can transmit filter context completely **outside** the current file to entirely different reports.

The Programmable Button Engine

Available Types

- ← Left arrow
- Right arrow
- ↶ Reset
- ↶ Back
- i Information
- ? Help
- 💬 Q&A
- 🔖 Bookmark
- Blank
- 🗃️ Navigator >



Without an action, a button is just a fancy text box. The **Action** engine defines functionality.

Format >>

Search

Button

General

...

▼ Action

On

Type

Drill through

▼

Bookmark

None

▼

Destination

Drill Through Page

▼

fx

Web URL

fx

Transparent UX Case Study

Scenario: Users may not know to right-click a visual to drill through.

Solution: Insert > Blank Button. Set Action Type to 'Drill through'. Point destination to the correct page.

▼ Action

On

Type

Drill through

▼

Bookmark

None

▼

Destination

Drill Through Page

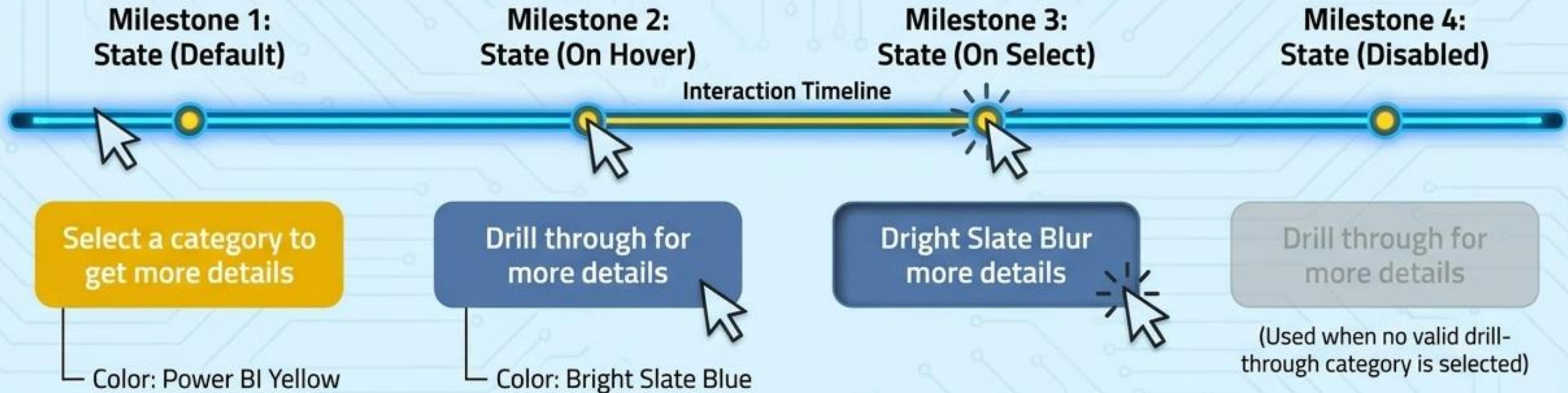
▼

fx

Web URL

fx

Button States & Micro-Interactions



Architectural Takeaway: Leveraging states allows developers to design in layers of functionality and micro-instruction directly on the canvas.

Automated Navigation Arrays

The Output (Front-End UI)

Page 1Page 2Page 3Page 4Page 5Page 6Page 7

Sales for Products in the Beverages category

Auto-Highlighting: The navigator automatically detects the current page and highlights the corresponding tab.

The Configuration Engine (Back-End Settings)

Format

Filters

Search

VisualGeneral...

> Style

Pages

Show hidden pages

Show tooltip pages

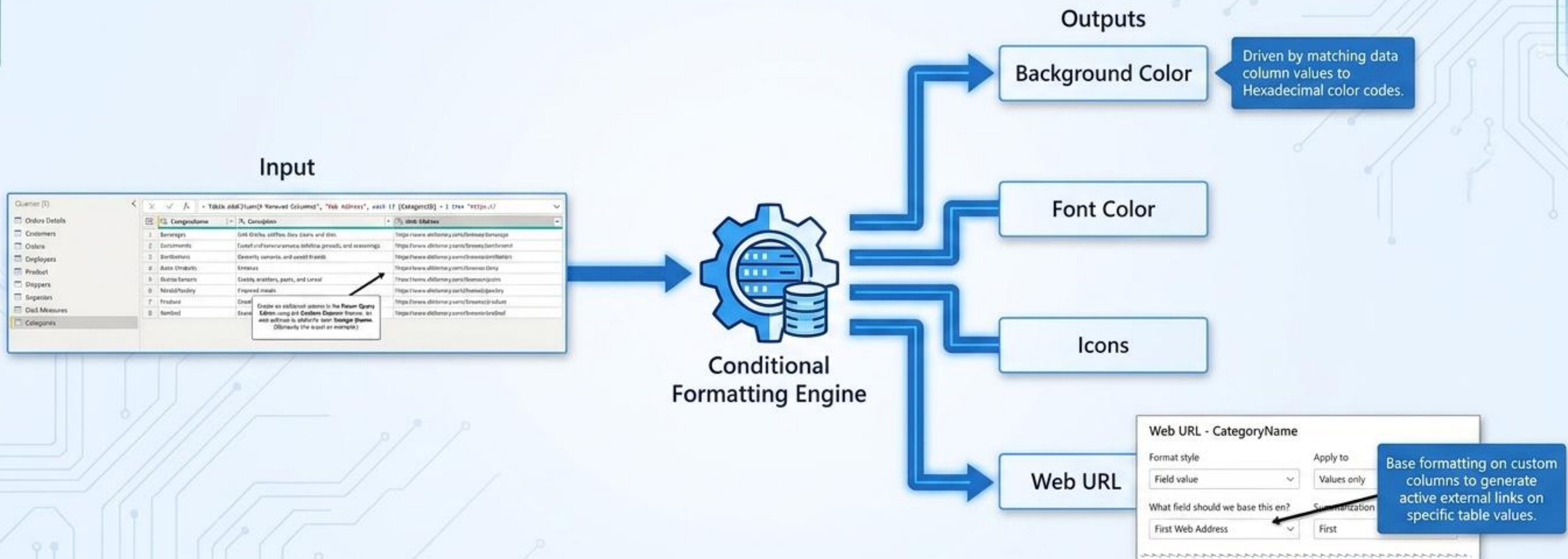
Visibility Controls: Toggles explicitly control whether 'Hidden pages' or 'Tooltip pages' render in the navigation menu.

Seamless scaling. Copies across pages instantly. Replaces manual bookmarking arrays.

The Navigational Mechanisms Matrix

MECHANISM	USER ACTION REQUIRED	MAINTAINS FILTER CONTEXT?	DESTINATION
Drill-Through Action	Right-click visual (or dedicated button) 	YES (Packs and sends filters.) 	Target Page (Internal or Cross-Report)
Page Navigator	Left-click top menu tab 	NO (Pure page navigation.) 	Specific internal report page
Web URL	Left-click icon/text 	NO (unless built into custom URL query).	External Web Address 

Conditional Formatting: The Rules Engine

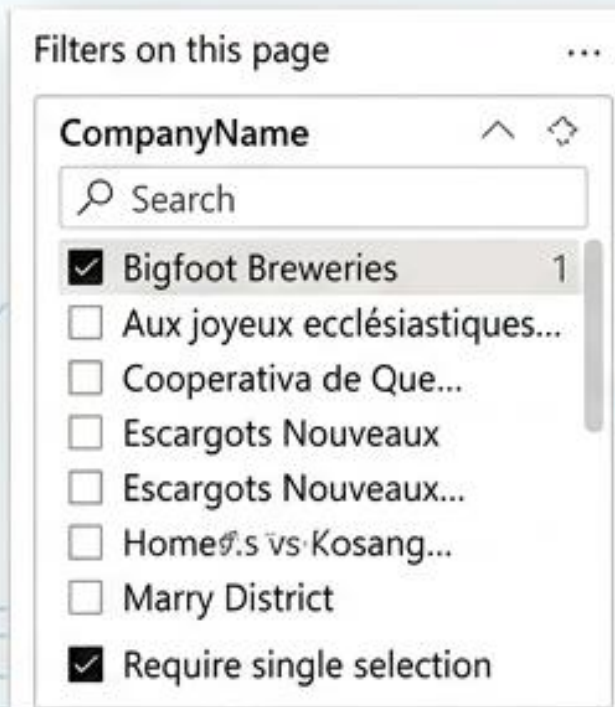


Anatomy of a Dynamic DAX Measure

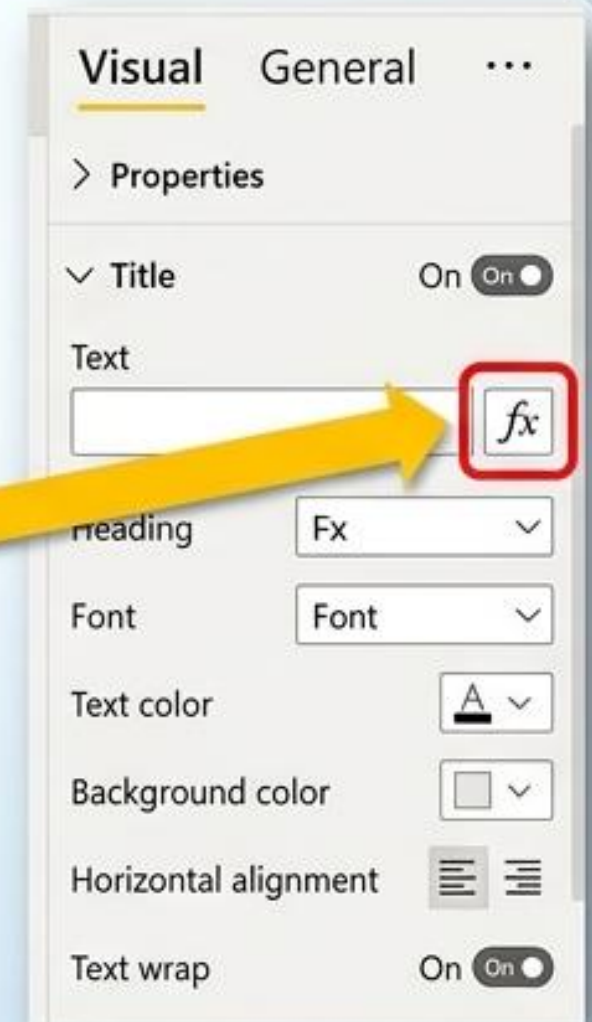
```
Title = "All Sales by Country for " & SELECTEDVALUE('Suppliers'[CompanyName])
```

Static text string.

Concatenation operator.



The UI Bind: Anywhere the Fx symbol appears, a measure can dictate the visual format. Clear the default static text and point the field directly to the dynamic DAX measure.



The Output Diagnostic Table

Standard Desktop Reports



- **Canvas Size:** Dynamic / Screen-bound.
- **Data Volume:** Scrollable visuals.
- **Layout Precision:** Relative / Responsive.
- **Primary Use Case:** Interactive Data Exploration.

Paginated Reports



- **Canvas Size:** Fixed / Paper dimensions.
- **Data Volume:** Renders all rows simultaneously (e.g., full inventory logs).
- **Layout Precision:** Pixel-perfect placement (e.g., pre-printed stationary/forms).
- **Primary Use Case:** Invoices, ledgers, and formal static reporting.

Built using the free Power BI Report Builder.

Published to the Service via Premium capacity workspace.

Pro-Tip: You can embed a paginated report directly into a standard interactive report using the dedicated paginated visual.